Group Work Contribution Manager

User and System Requirements

This document specifies a contribution tracking application for a group of six students. Members record their work, rate their own contributions from 0 to 5, review and dispute ratings, and obtain contribution percentages for each coursework submission. User requirements describe user needs; system requirements define the application behaviour and quality targets.

1 Calculation and approval rules

Each student submits a 0–5 integer rating and a short explanation for every completed task. A rating of 0 means no contribution; a missing rating remains pending.

Tasks should be roughly similar in size. This version does not use task weights. Ratings do not have to total five for a task.

All six members must accept the complete set of ratings before a task is finalised. Open disputes prevent finalisation. Only finalised tasks count towards percentages.

Contribution percentage = (member total approved score / combined approved score of all six members) × 100.

Calculations are separate for Coursework 1 and Coursework 2. If the combined score is zero, the system displays “Not yet calculated”. Percentages describe contributions and are not academic marks.

Rating scale

Score	Meaning
0	No contribution to the task
1	Minor assistance
2	Completed a limited part
3	Made a substantial contribution
4	Completed most of the work
5	Completed the entire task independently

The approval rules and performance targets below are proposed project design choices.

2 User requirements

ID	User requirement
UR01	Students need to register, sign in and sign out securely.
UR02	The lecturer will need to add students to a group.
UR03	Students need to be able to see who is in their group.
UR04	Lecturer should be able to structure the project based on the coursework specification (eg. Chapter 1 : Problem specification, Task 1: Process of gathering user requirements)
UR05	Students should be able to view tasks, including descriptions, chapters and deadlines.
UR06	Students need to track whether tasks are not started, in progress or completed.

ID	User requirement
UR07	Students need to record their contributions to completed tasks and provide supporting evidence.
UR08	Each student needs to rate their own contribution to every completed task from 0–5.
UR09	Students should be able to give clear guidance explaining each rating.
UR10	Lecturers should be able to view and review the group's submitted contributions and ratings.
UR11	Students should be able to dispute a contribution or rating and explain their disagreement.
UR12	Students should be able to revise their own submissions and agree final ratings.
UR13	Students should be able to view missing ratings, pending approvals, and unresolved disputes.
UR15	Students and Lecturers should receive notifications when an action (disputes, marks altered, comments added etc.) is required.
UR16	Students and lecturers should be able to view a history of changes to contributions, ratings, and disputes.

3 Calculation example

Suppose the six members have approved totals of 4, 4, 5, 4, 4 and 3. The combined group score is 24.

Member	Approved score	Contribution percentage
A	4	16.67%
B	4	16.67%
C	5	20.83%
D	4	16.67%
E	4	16.67%
F	3	12.50%

For member C: (5 / 24) × 100 = 20.83%. The displayed percentages total 100.01% because of rounding. If all six members have equal positive scores, each receives approximately 16.67%.